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Effect of epoxy resin addition on the acoustic impedance, microstructure, dielectric and piezoelectric properties of 1–3 connectivity lead-free barium zirconate titanate ceramic cement-based composites

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ABSTRACT

In this work, 1–3 connectivity barium zirconate titanate ceramic cement-based composites were fabricated using Portland cement and epoxy resin as the matrix. Barium zirconate titanate (BZT) of 40–60 % by volume was used while epoxy was used with cement at 0–7% by volume. Dielectric and piezoelectric properties, and other properties such as acoustic impedance, density and microstructure were investigated. It was found that epoxy resin can be used in combination with BZT to achieve a suitable acoustic impedance value (9–11 × 10^6 kg/m·s^2) matching that of concrete for structural health monitoring application. Thus, when epoxy resin was used at 7 %, BZT volume can be increased to 60 % where the highest d_{33} value of 93 pC/N was found and remain within the acoustic matching range. In addition, when epoxy resin was increased, both piezoelectric charge coefficient (d_{33}) and piezoelectric voltage coefficient (g_{33}) were also found to increase. This is likely due to the lower porosity thus denser matrix when epoxy resin was used in addition to cement.

1. Introduction

Piezoelectric materials can convert a mechanical vibration to an electrical signal and vice versa, which can be used as sensors and transducers. However, due to the mismatching to concrete, piezoelectric ceramic-cement composites have been developed [1–3] to match concrete for structural health monitoring (SHM) application to inspect and detect structural changes (e.g., stress, strain) and damage of infrastructures in real-time [4–8]. These piezoelectric ceramic-cement based composites can be embedded as sensors to detect crack propagation in mortar samples under cubic splitting force enables the in-place stress monitoring of mortar and concrete beams under constant compression [9–11].

Although lead based ceramic have been used in piezoelectric ceramic cement-based composites [1–3,12–15] but in order to overcome detrimental effect on the environment and human health, lead-free piezoelectric materials have emerged as a sustainable alternative to conventional lead-based piezoelectric materials [16–18]. Lead-free barium titanate-based ceramics such as barium titanate ($BaTiO_3$, BT) ceramic and barium zirconate titanate ($BaZr_xTi_{1-x}O_3$, BZT) ceramic have also been used [19–22]. BZT ceramic is a modified BT ceramic where substituting Ti by Zr of 0.05 was found to have improved piezoelectric properties [23–25].

A variety of connectivity patterns such as 0–3 and 1–3 may be used to produce the piezoelectric ceramic cement-based (PCC) composites. Normally the 1–3 in the composite defines the connectivity of each phase where the first digit number represents connectivity of the active phase such as piezoelectric ceramic and the second digit number represents connectivity of passive phase which is cement matrix in this case. Therefore, 1–3 composite in this investigation has BZT ceramic phase as self-connected in one dimension and the PC matrix phase as self-connected in three dimensions [26].

Although, the advantage of 0–3 composite is easier fabrication than 1–3 composite but these composites are more difficult to pole due to the non-connectivity of the dispersed phase. To overcome this problem, a polymer third phase such as polyvinylidene fluoride (PVDF) [27–31] and epoxy resin [32,33] may be added in these 0–3 composites. The optimum of acoustic impedance (Zc) and piezoelectric voltage coefficient g_{33} can be observed at 5 vol% PVDF [30] and 3 vol% epoxy resin [33] with same 50 vol% BZT. Nonetheless, the 1–3 composites have also been investigated [34–37] since the piezoelectric phase in the 1–3 composites is parallel with the electric field polarization thus the

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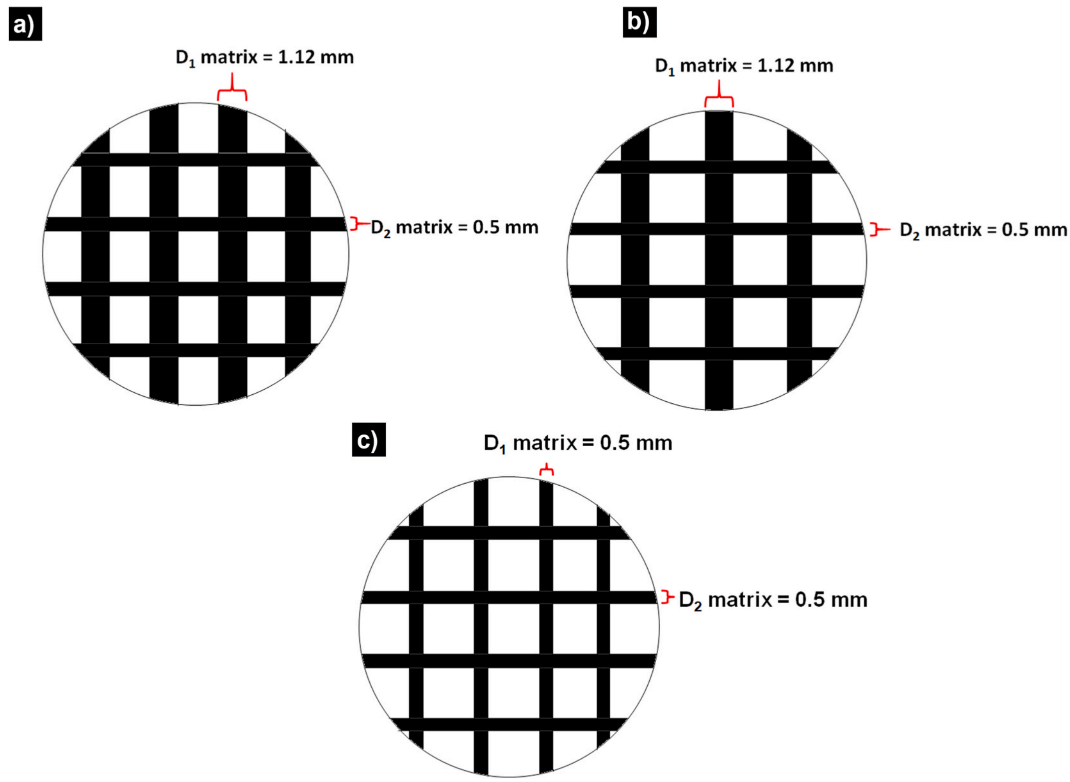

Fig. 1. Schematic diagram of 1–3 BZT cement-based composites using BZT ceramic of a) 40 ± 2 vol%, b) 50 ± 2 vol% and c) 60 ± 2 vol%.

ceramic is poled first without poling problem as in 0–3 connectivity, resulting to piezoelectric properties being enhanced. The highest d_{33} value of 1–3 BZT-PC composite reported by Potong et al. [36] is 133 pC/N at 70 vol% BZT which is 5 times higher than 0–3 BZT-PC composite (d_{33} = 27 pC/N) at same BZT content [21].

Some advantages of epoxy resin are good adhesion with low shrinkage and high electrical resistivity. Thus, the epoxy resin may be used as matrix phase in modified cementitious materials for applications in electronic devices [38–40]. In addition, epoxy resin has a significantly lower Z_c value compared to other materials: therefore, acoustic matching can be tailored with increasing piezoelectric ceramic volume while maintaining Z_c close to concrete.

In this work, composites of 1–3 connectivity of barium zirconate titanate ceramic with cement and epoxy resin were fabricated. BZT of 40–60 % by volume and epoxy of 0–7% by volume were used. Dielectric and piezoelectric were investigated. Other properties such as acoustic impedance, density and microstructure were also investigated.

2. Materials and methods

2.1. BZT preparation

BZT ($BaZr_xTi_{1-x}O_3$, BZT) powder was prepared where x = 0.05 [30] using the mixed oxide method [15] by calcining barium carbonate (purity ≥99.0 %, Sigma-Aldrich, Italy), zirconium dioxide (purity = 99.0 %, Riedel de Haën, France) and titanium dioxide (purity ≥99.0 %, Sigma-Aldrich, Germany) at 1200 °C for 2 h. Then, BZT powder was pressed into a disc shape of ~15 mm diameter and a thickness of ~1.5 mm. The applied pressure for BZT discs fabrication is ~85 MPa. Thereafter, BZT discs were sintered at 1450 °C for 2 h.

2.2. Fabrication of 1–3 BZT cement-based composites with epoxy resin

BZT composition of 40, 50, and 60 vol% BZT were used (BZT40, BZT50, and BZT60) and epoxy resin (ER) was added at 0, 3, 5, and 7 vol % with the cement paste. First, BZT discs (dimension ~12 mm and thickness ~1.5 mm) were poled using an electric field of 1.5 kV/mm at 50 °C for 30 min in a silicone oil chamber. By using a cut and fill method, poled BZT ceramic was cut using a diamond saw in a parallel direction along with the polarization axis. The designed schematic diagram of 1–3 BZT cement-based composites is shown in Fig. 1. Mixtures of Portland cement powder (ASTM Type I), water, epoxy resin and hardener were used as the matrix phase in the composite. Commercial epoxy resin (Epofix resin, Struers, Ballerup, Denmark) is formulated using bisphenol-A-(epichlorhydrin), while the commercial hardener (Epofix hardener, Struners, Ballerup, Denmark) consists of Triethylenetetramine. The water to cement ratio (w/c) used is 0.35 and the volume ratio of epoxy resin and hardener is 7.5. The mixture was filled into the cut areas of the BZT ceramic. The samples were then cured in a controlled chamber at 60 °C for 5 days at 98 %RH.

2.3. Acoustic impedance, microstructure, dielectric and piezoelectric measurements

Acoustic impedance (Z_c) was calculated by multiplying the density (ρ_c) and acoustic velocity (v_c) of the composites. The v_c was measured using an ultrasonic thickness meter (TM-8812C). An optical microscope and a scanning electron microscope (SEM-5910LV) was used to investigate the microstructure and interface, and energy dispersive X-ray from the equipped SEM was used to detect elements of the composites.

The ε_{33} at frequency of 1 kHz of the composites can be calculated from the measured capacitance (C) and tanδ values using a LCR meter (TH2819A, Tounghui) from equation (1) as shown:

$$\varepsilon_{33} = \frac{Ct}{\varepsilon_0 A} \quad (1)$$

where t represents the thickness of the sample (m), ε_0 (8.854 pF/m), is the relative permittivity in free space and A is the electrode area of the sample (m^2).

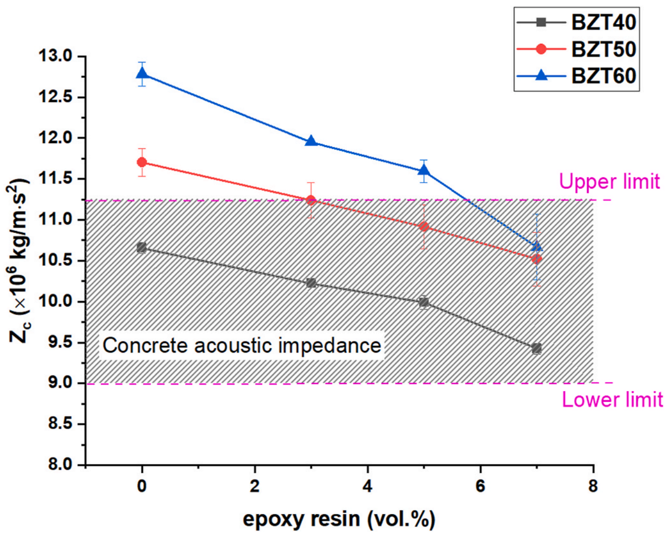

Fig. 2. Effect of epoxy resin content on Z_c value of 1–3 BZT cement-based composite with various epoxy resin addition.

The d_{33} meter (S5865, KCF technologies) was used to obtain the piezoelectric charge coefficient (d_{33}). The piezoelectric voltage coefficient (g_{33}) was calculated using equation (2) as follows:

$$g_{33} = \frac{d_{33}}{\varepsilon_r \varepsilon_0} \tag{2}$$

3. Results and discussion

3.1. Acoustic impedance

The acoustic impedance of 1–3 BZT-cement composites using BZT content from 40 to 60 % and epoxy resin from 0 to 7% is shown in Fig. 2. A higher Z_c value of the composites can be observed in composites with higher BZT volume. This is because the Z_c value of BZT (25.7×10^6 kg/m·s^2) is higher than other phases in the composite (Z_c of hydrated cement $\sim 5.3 \times 10^6$ kg/m·s^2 [1], Z_c of epoxy resin $= 2.99 \times 10^6$ kg/m·s^2). The effect of epoxy resin on the acoustic impedance of 1–3 BZT -cement composites show that Z_c value can be seen to be lower when epoxy resin content was increased for all BZT composites. For example, the composites with 40 vol% BZT and epoxy resin content up to 7 %, although reduces from $\sim 10.7 \times 10^6$ kg/m·s^2 to $\sim 9.4 \times 10^6$ kg/m·s^2, the acoustic impedance still matches with that of concrete ($Z_c \sim 9 \times 10^6$ kg/m·s^2 to 11.26×10^6 kg/m·s^2 [1,41]). For 50 vol% BZT composites with no epoxy resin, Z_c value ($\sim 11.7 \times 10^6$ kg/m·s^2) is marginally higher than the acoustic impedance upper limit of concrete. However, the composites with 3 vol% to 7 vol% epoxy resins are found to be within the acoustic range. For 60 vol% BZT composites, 0–5% vol% epoxy resin are found to be outside the range but with 7 vol% epoxy resin in the composite, Z_c value is in a range of acoustic impedance of concrete. Therefore, the benefit is seen when increasing epoxy resin content in the composites with 50 vol% and 60 vol% BZT where the use of epoxy resin would lead to a decrease in acoustic impedance so that the acoustic impedance becomes closer to the acoustic impedance of concrete when higher than 40 % BZT volume is used.

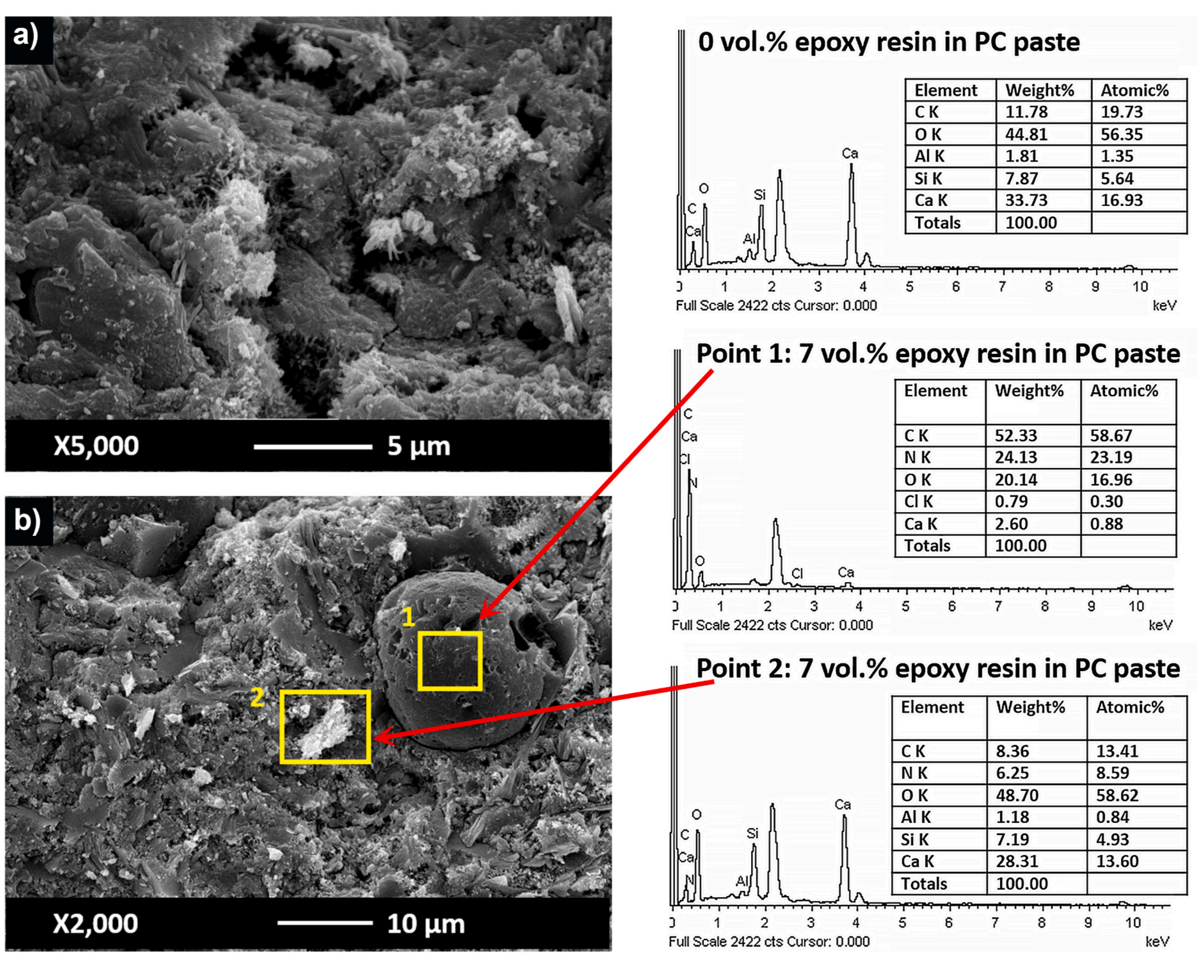

Fig. 3. SEM micrograph images and EDS spectra of the cement matrix at w/c = 0.35 with epoxy resin at a) 0 vol% and b) 7 vol%.

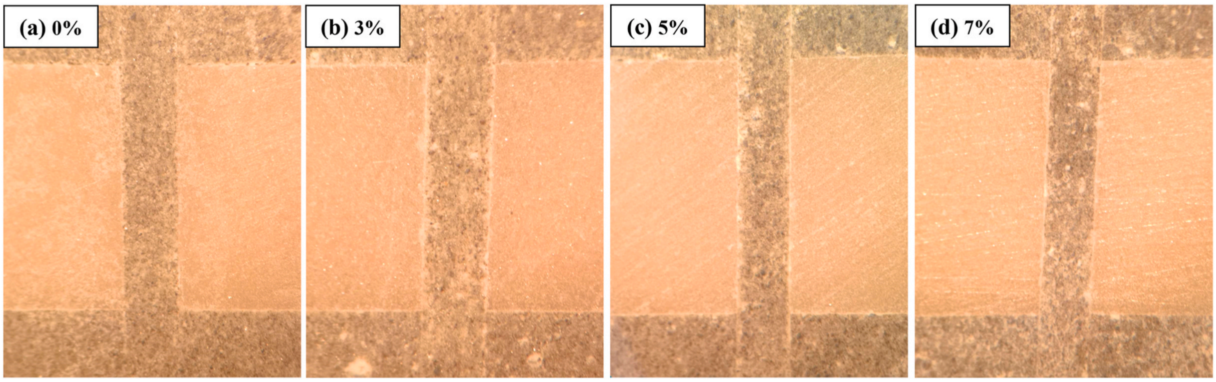

Fig. 4. Optical microscopy images of 1–3 BZT cement-based composites the interface with epoxy resin at a) 0 vol%, b) 3 vol%, c) 5 vol% and d) 7 vol%.

3.2. Microstructure and interface

Microstructure investigation was carried out using a scanning electron microscope (SEM) images and energy dispersive X-ray spectrum (EDX) on the matrix in the composites are shown in Fig. 3. Fig. 3(a) shows the microstructure of Portland cement paste with no epoxy resin where calcium silicate hydrate (CSH) can be seen. Ca, Si, C, Al, and O elements were detected as the main elements of hydration products from cement by the EDX spectrum. Fig. 3(b) shows the SEM micrograph of cement and epoxy resin matrix where the EDX spectrum (point 1) identifies the epoxy resin, where detected elements are nitrogen and carbon. In spectrum 2, elements detected are Ca, Si, C, Al, O and N elements, which illustrates the combination phases between the epoxy resin and hydration products of hydrated cement. Moreover, the interfaces of the composites are shown in Fig. 4 using an optical microscope and Fig. 5 using SEM. The optical microscopy clearly shows good binding in a larger scale dimension for all composites with epoxy resin (3%–7%) and without epoxy resin (0 %). In the matrix of cement and epoxy a rather bright dots of epoxy resin can be seen on the surface. For SEM, the images of the 1–3 BZT cement-based composite for both composites with 0 vol% and 7 vol% epoxy resins, the BZT ceramic can be seen to be continuously connected in one dimension. The interfacial zone can also be clearly observed between cement matrix paste (dark area) and BZT phase (bright area). For comparison, EDX spectrum from the cement matrix phase of the composite with epoxy resin shows nitrogen element while in sample without epoxy resin, nitrogen element was not detected, thus corresponds to the EDX spectrum of PC paste with w/c = 0.35 (Fig. 3).

3.3. Porosity results

Fig. 6 shows the porosity measurement results of the composites. The porosity can be seen to be highest in the composites with BZT vol. content of 40 % where the porosity values are in the range of 8.5–9%. This is higher than the porosity range of 7–8% for 60 % BZT composites. This is because of the higher cement/epoxy matrix content which is more porous than the BZT ceramic. Moreover, the effect of epoxy resin can also be seen, where it is found that for all BZT composites, the porosity is shown to reduce with increasing epoxy resin. In the 1–3 connectivity composite, it is believed that the use of epoxy resin was found to fill the pores of the cement matrix thus lead to a lower porosity value.

3.4. Dielectric properties

Figs. 7 and 8 show the dielectric constant and dielectric loss respectively. The ε_{33} value can be seen to increase but tanδ value is seen to decrease with increasing BZT content at the same epoxy resin content. At 7 vol% epoxy resin, the ε_{33} value of composites with BZT content from 40 vol% to 60 vol% increased from 464.79 to 636.04, showing an increase of 37 %. While the tanδ value decreased from 0.07 to 0.04 and the tanδ value was lowest for composites with 60 vol % BZT/7 vol% epoxy resin. A parallel model [22] is also used to compare to the real measurements (Figs. 9–12) where it can be seen that the parallel model is slightly higher than that of the values taken from actual measurements. Moreover, for all BZT content in the composites, the ε_{33} and tanδ values can be seen to decrease with increasing epoxy resin content. At 50 vol% BZT content, the ε_{33} value of composites with epoxy resin vol. Content from 0 to 7 % decreased from 622.16 to 521.10 and tanδ decreased from 0.08 to 0.05. This is because the dielectric constant of BZT (ε_r = 1563) is higher than hydrated cement (ε_r = 100) and epoxy resin (ε_r = 6). The much lower tanδ = 0.03 [42] of epoxy resin in which using epoxy resin replacing the cement phase lead to a direct decrease in the overall dielectric loss of the composite since cement is a porous inorganic material and has many weak conducting ions [43].

3.5. Piezoelectric properties

Fig. 13 shows the effect of epoxy resin content on the d_{33} value of 1–3 BZT-cement composites. As the BZT content increases, the d_{33} value of composite showed a corresponding increase, primarily due to the higher amount of the piezoelectric phase within the composite. Specifically, when epoxy resin content was 7 vol%, the d_{33} value demonstrated a notable increase from 76.2 pC/N to 93 pC/N as the BZT content rose from 40 vol% to 60 vol%. Furthermore, the highest d_{33} value (93 pC/N), was observed when the composite consisted of 60 % BZT and 7 vol% epoxy resin, while d_{33} of pure BZT is 235 pC/N.

In the composite with varying BZT content, the d_{33} value increased as the epoxy resin content was increased, as shown in Fig. 13. For instance, at 50 vol% BZT, the mean d_{33} values increased from 79 pC/N to 83 pC/N (Table 1) as the epoxy resin vol. Increased from 0 % to 7 %. This finding agrees with the results reported by Xu et al. [35] on the 1–3 PZT cement-polymer composites.

It is noted that the d_{33} values of 1–3 BZT cement composites with epoxy resin is noticeably higher and the trend observed is different to that of 0–3 BZT cement composites with epoxy resin [33], where the d_{33} value of 0-3 connectivity composites decreased with increasing epoxy resin content. This difference is likely to be due to different type of the sample connectivity, the poling process and the poling mechanism. The ceramic sample of the 1–3 piezoelectric cement-based composite is poled prior to the fabrication (Poling sample is pure BZT ceramic). The use of epoxy resin in the composite reduces the presence of pores within cement and at the interface between cement and BZT, which likely to improve transmitting force to the BZT phase, when apply oscillating force during the d_{33} measurement and results in improve d_{33} value. This is due to that porosity of the 1–3 composites decreased with increasing epoxy resin content as shown in Fig. 6.

While for the 0–3 composite, the poling of sample is slightly more

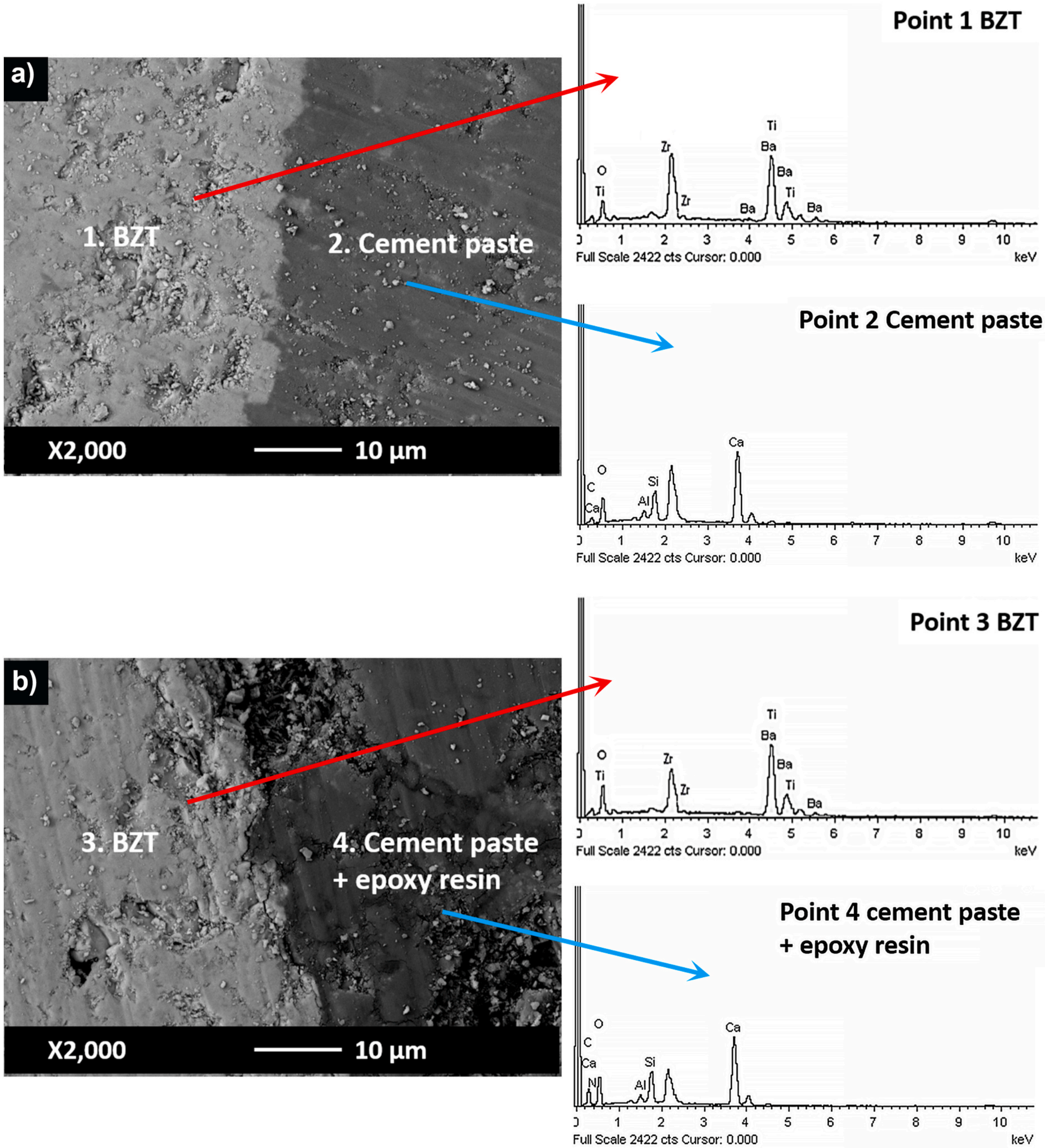

Fig. 5. SEM micrograph images of 1–3 BZT cement-based composites the interface with epoxy resin at a) 0 vol%, b) 3 vol%, c) 5 vol% and d) 7 vol%.

difficult with epoxy addition in the 0–3 composite as the whole composite was poled together. The d_{33} values decreased when increasing epoxy resin due to that the epoxy resin acts as an insulator phase that may block or surround the piezoelectric particles as it was added in its resin form, which may result in a more resistivity electric field flow to BZT particles and unsaturated polarization.

The effect of varying epoxy resin content on g_{33} value of 1–3 BZT-cement-based composite is shown in Fig. 14. An increase in BZT content led to a decrease in the g_{33} value. At 7 vol% epoxy resin, g_{33} value decreased from 18.5×10^{-3} V m/N to 16.5×10^{-3} V m/N when the BZT content ranged from 40 vol% to 60 vol%. This decrease is primarily attributed to the fact that the increase in the ε_{33} value of the 1–3 composite is more than the increase in the d_{33} value. For example, at 7 vol% epoxy resin, the increasing rate ε_{33} was 37 % when varying BZT from 40 to 60 vol%, while the increasing rate of d_{33} value was less at 22 %. These findings align with previous research by Xu et al. [35], Potong et al. [36], and Rianyoi et al. [44]. Even though, d_{33} value of the composite with 40 vol% BZT and 5 vol% epoxy resin (75 pC/N) is lower than that of BZT ceramic (240 pC/N), the g_{33} value of the composite (17.0×10^{-3} V m/N) is equal to BZT ceramic (17.0×10^{-3} V m/N).

The result indicates that an increase in epoxy resin content leads to a rise in the g_{33} value for all composites using 40 % BZT to 60 % BZT (Fig. 14). At 50 vol% BZT, the g_{33} value of the composite with epoxy resin ranging from 0 vol% to 7 vol%, increased from 14.4×10^{-3} V m/N to 18.1×10^{-3} V m/N. This result aligns with the results reported by Xu et al. [35] for 1–3 PZT cement-polymer composite.

4. Conclusions

When epoxy resin was increased, both piezoelectric charge

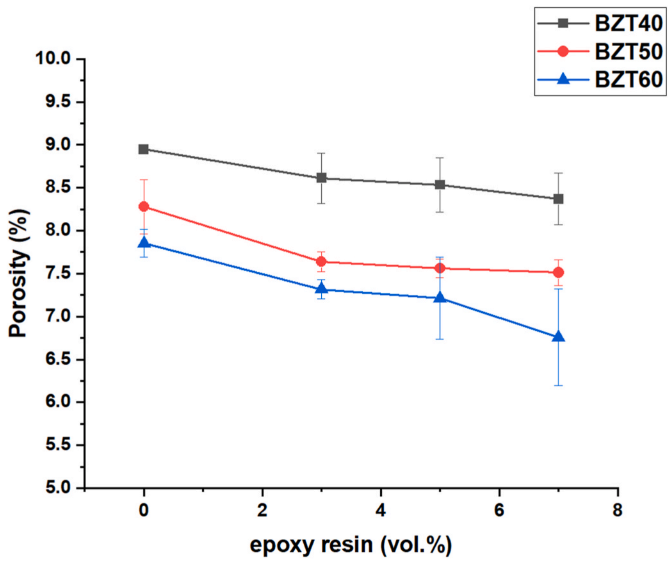

Fig. 6. Effect of epoxy resin content on porosity of 1–3 BZT cement-based composite with 40 vol% to 60 vol%. BZT and 0 vol% to 7 vol% epoxy resin.

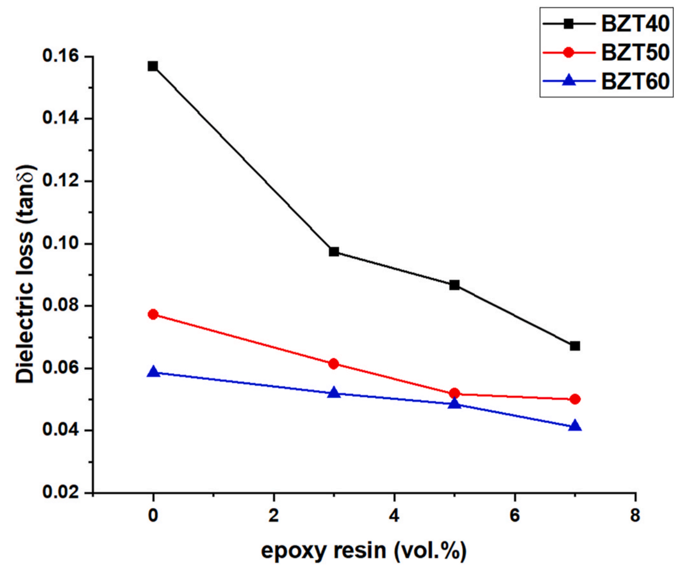

Fig. 8. The effect of the epoxy resin content on tanδ of 1–3 BZT cement-based composites with epoxy resin addition at a frequency of 1 kHz.

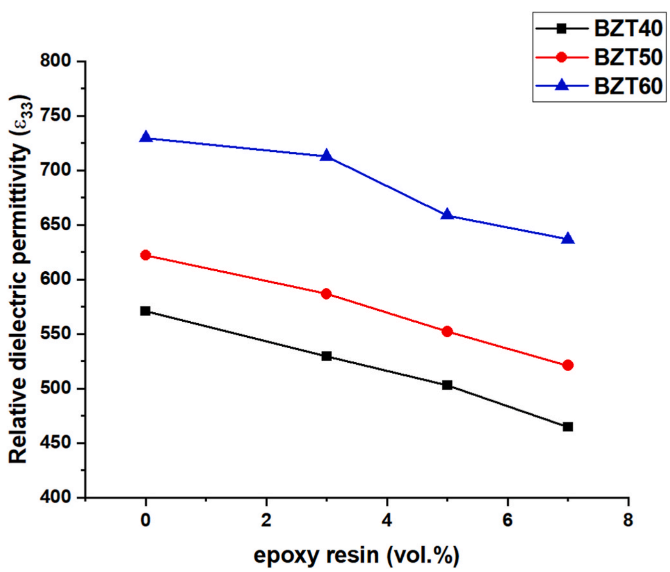

Fig. 7. The effect of the epoxy resin content on ε_{33} of 1–3 BZT cement-based composites with epoxy resin addition at a frequency of 1 kHz.

coefficient (d_{33}) and piezoelectric voltage coefficient (g_{33}) were also found to increase. This is likely due to the lower porosity thus denser matrix when epoxy resin was used in addition to cement. The effect of epoxy resin on 1–3 BZT composites are also shown to reduce the acoustic impedance, and suitable percentage can be designed for ideal acoustic impedance matching with concrete achieving values within the range of 9–11 $\times$ 10^6 kg/m·s^2 for structural health monitoring purposes. The matching was found in composites with 40 vol% BZT (0–7% epoxy resin), 50 % vol. BZT (3–7% epoxy resin) and 60 vol% BZT (7 % Epoxy resin). Thus, when epoxy resin was used at 7 %, BZT volume can be increased to 60 % where the highest d_{33} value of 93 pC/N was found and remain within the acoustic matching range. The d_{33} values of 1–3 connectivity composite with epoxy are much higher than those of previously reported values of 0–3 composites with epoxy resin.

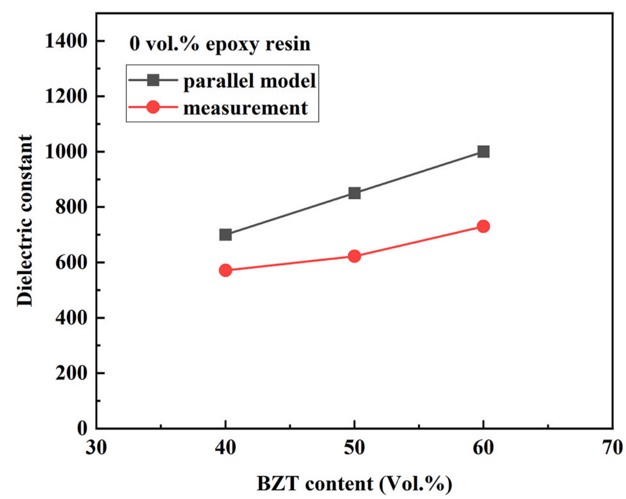

Fig. 9. Comparison between parallel model to the dielectric constant measurements of 1–3 BZT cement-based composites with no epoxy resin addition.

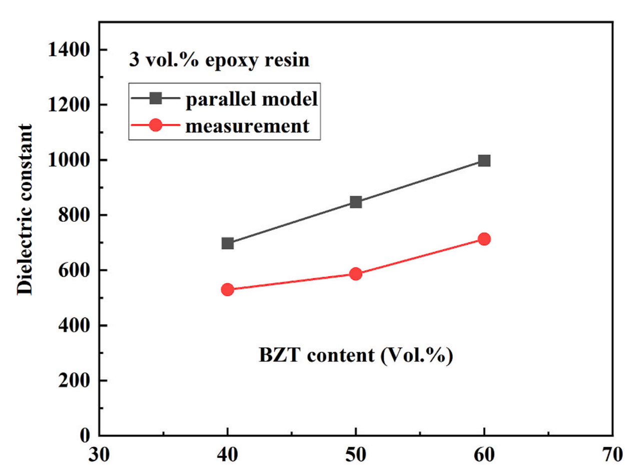

Fig. 10. Comparison between parallel model to the dielectric constant measurements of 1–3 BZT cement-based composites with 3 % epoxy resin addition.

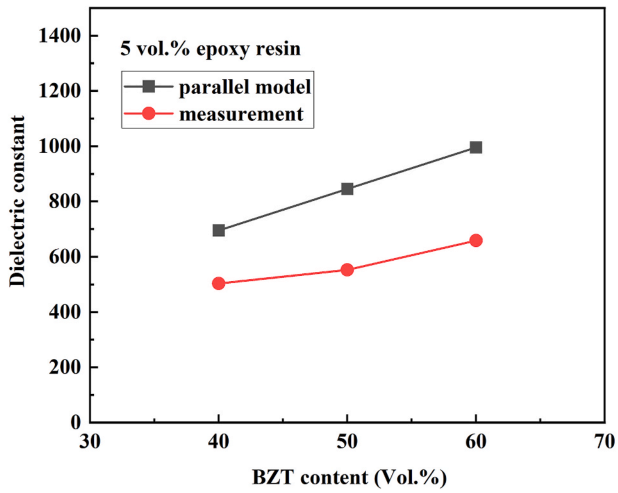

Fig. 11. Comparison between parallel model to the dielectric constant measurements of 1–3 BZT cement-based composites with 5 % epoxy resin addition.

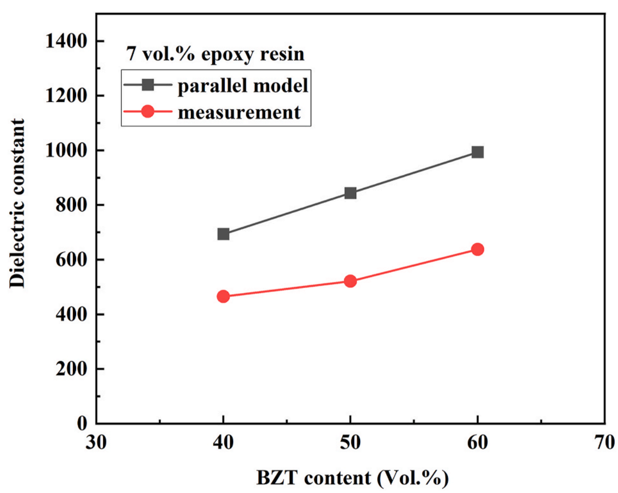

Fig. 12. Comparison between parallel model to the dielectric constant measurements of 1–3 BZT cement-based composites with 7 % epoxy resin addition.

CRediT authorship contribution statement

Thanyapon Wittinanon: Formal analysis, Investigation, Writing – original draft. **Rattiyakorn Rianyoi:** Formal analysis, Supervision, Writing – original draft, Writing – review & editing. **Ruamporn Potong:** Investigation, Writing – original draft, Writing – review & editing, Formal analysis. **Arnon Chaipanich:** Conceptualization, Formal analysis, Investigation, Methodology, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests.

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2. Thailand Science Research and Innovation (TSRI) formerly known as the Thailand Research Fund (TRF) for the Royal Golden Jubilee PhD Program Grant: (Grant No. PHD/0179/2557) to Thanyapon Wittinanon (Prof. Dr. Arnon Chaipanich: advisor)

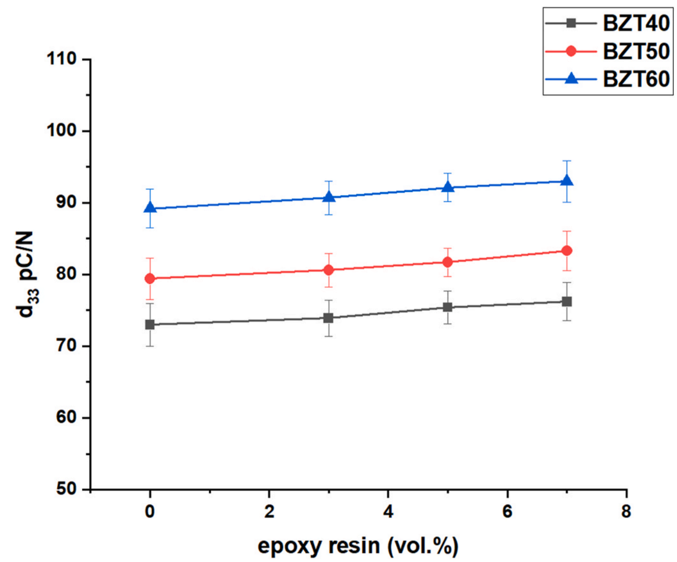

Fig. 13. The effect of the epoxy resin content on d_{33} of 1–3 BZT cement-based composites with epoxy resin addition.

Table 1
The d_{33} values of 0–3 and 1–3 BZT-PC composites at 50 vol% BZT with 0–7 vol% epoxy resin.

Epoxy resin (vol%)	Mean d_{33} (pC/N)	
	1-3 composite (this work)	0-3 composite [33]
0	79	16.5
3	81	14.4
5	82	13.3
7	83	12.6

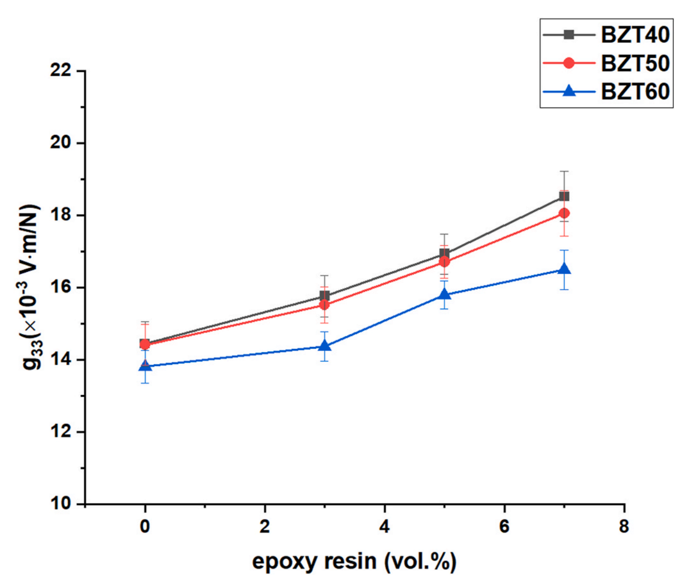

Fig. 14. The effect of the epoxy resin content on g_{33} of 1–3 BZT cement-based composites with epoxy resin addition.

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